Comparative Performance Analysis of FinFET and Gate-All-Around Dual Nanosheet FET

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***Abstract*—This coursework report presents a comparative DC performance analysis of the FinFET and the dual nanosheet gate-all-around FET (GAAFET) simulated using Sentaurus TCAD. The effect of fin corner rounding on electrostatic integrity and current conduction is investigated, showing that rounding degrades most performance metrics due to reduced gate-controlled area and weakened short-channel immunity. A dual nanosheet GAAFET design is presented, employing a TiN metal gate with full channel wrap-around, demonstrates superior electrostatic control, achieving near-ideal subthreshold swing=60.13 mV/dec, DIBL=28.4 mV/V, and I_{on}/I_{off}=1.08×10^{10}, highlighting the fundamental advantages of GAA architectures for continued transistor scaling at nm technology nodes.**

I. Device Structures & Simulation Setup

The simulations were performed in Sentaurus TCAD using a hydrodynamic transport model, solving Poisson and carrier continuity equations self-consistently with electron temperature to capture high-field effects in nanoscale devices.

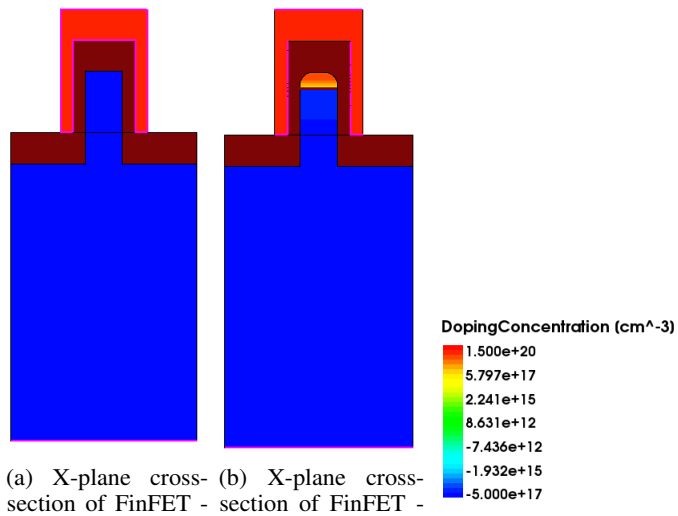
The simulations were carried out using a quasistationary bias ramp to ensure numerical stability and convergence. For transfer characteristics in the linear region, the drain voltage was first ramped from 0 V to 10 mV, followed by a gate voltage sweep from -0.5 V to 2 V to capture subthreshold through strong inversion behaviour. For saturation operation, the drain voltage was instead ramped to 1.5 V prior to sweeping the gate voltage over the same range, ensuring the device operated in the high-field regime. For output characteristics, the drain voltage was swept from 0 V to 1.4 V at fixed gate bias, 2V after establishing the operating point, allowing the transition from linear to saturation regions to be captured.

A. FinFET with Rectangular Fin

The fin (channel) and substrate are p-type, doped with Boron at 5e17 cm³. Doping of source, drain and gate is 5e19 cm³. The source, drain and gate are doped with Arsenic (donor impurity) at 5e19cm³. The fin is formed out of the same silicon as the substrate, but its lighter doping and thinner geometry are deliberately engineered for strong gate control.

B. FinFET with Rounded Fin

The rounded fin structure was implemented by applying a fillet operation to the top edges of the fin using `sdegeo:fillet`, introducing a curvature radius of $0.002\,\mu$m as shown in Figure 2. This modification mimics the rounding observed in practical fabrication due to etching and oxidation processes in STI-based FinFETs.



(a) X-plane cross-section of FinFET - Rectangular Fin (b) X-plane cross-section of FinFET - Rounded Fin (c) Doping Conc.

Fig. 1: FinFET structures

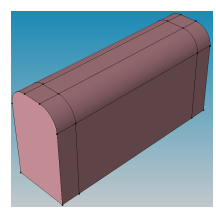


Fig. 2: Rounded Fin

As the flat top surface is removed, adjustment of the gate contact definition is required to ensure proper wrapping around the fin. To maintain consistent source/drain extension into the curved regions, an additional Gaussian analytical doping profile was applied from the fin top surface. Furthermore, local mesh refinement was introduced near the rounded edges to accurately capture the weakened electric field gradients due to the reduction of electric field crowding noted at sharp corners. All other structural and doping parameters were kept unchanged.

C. Dual Nanosheet GAAFET

The dual nanosheet GAAFET structure was implemented by replacing the fin with two vertically stacked, Si nanosheets, each with a thickness of 2 nm, surrounded by a 1 nm SiO_2 gate oxide, while a continuous gate of thickness 1 nm fully wraps

around both sheets, forming a gate-all-around configuration. This is shown in Figure 3,4,5. For electrical isolation, a base SiO_2 layer sits between the substrate and the source/drain/gate. The gate is also separated from the source/drain regions by free space to keep the struture simple. Ideally, the separation would be with an oxide.

Titanium nitride (TiN) is used as the metal gate to eliminate polysilicon depletion and due to its tunable work function (typically in the range ∼4.2–4.7 eV depending on process conditions), it is suitable for accurate engineering threshold voltage. The gate contact was assigned to the outer surface of the gate region such that the entire gate metal is treated as one electrode, providing uniform biasing around all channel surfaces. All other structural parameters were kept consistent with the baseline FinFET for fair comparison.

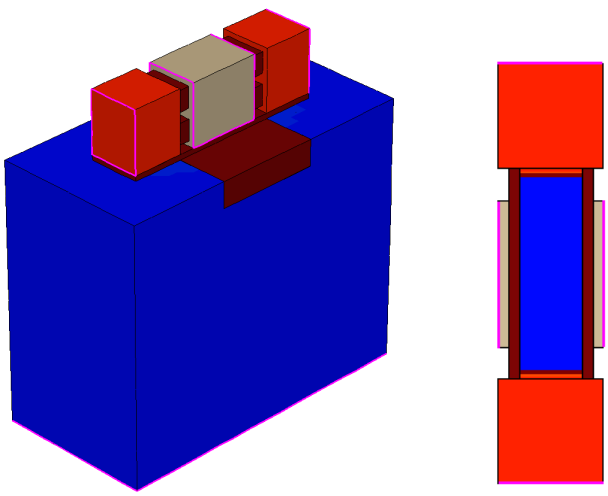


Fig. 3: Dual NS GAAFET structure and Z-plane cross-section

II. Electrical Characterisation Methodology for Parameter Extraction

The simulated I–V characteristics were exported from Sentaurus Visual as CSV files and processed in MATLAB for DC parameter extraction.

For V_{th} in the linear region, $g_m = \frac{dI_D}{dV_{GS}}$ in linear region was computed, and the gate voltage corresponding to its maximum

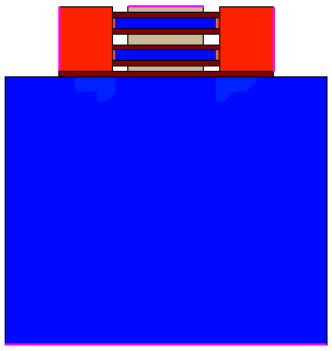


Fig. 4: Y-plane cross-section of NS GAAFET

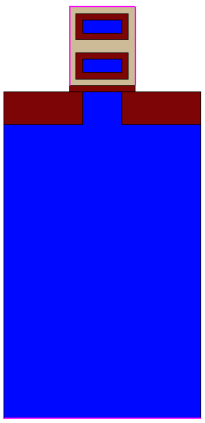


Fig. 5: X-plane cross-section of NS GAAFET

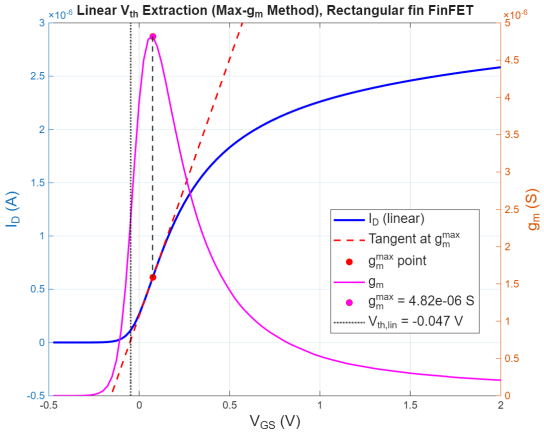


Fig. 6: Linear V_{th} Extraction

value, $V_{GS}\big|_{g_m^{\max}}$, was identified. A tangent line to the linear I_D–V_{GS} curve at this point was then extrapolated to intersect the V_{GS} axis, giving $V_{\text{th,lin}}$, shown in Figure 6. A correction of $V_{\text{DS}}/2$ is added because the extrapolation effectively references the threshold to the average channel potential, which is $V_{\text{DS}}/2$ above the source.

In saturation, V_{th} was extracted by approximating the most linear region (coincides with the point of maximum derivative) of the $\sqrt{I_D}$–V_{GS} curve in strong inversion and extrapolating it to $\sqrt{I_D} = 0$, where the intercept defines $V_{\text{th,sat}}$, shown in Figure 7.

The subthreshold swing (SS) was determined from the inverse slope of the $\log_{10}(I_D)$–V_{GS} curve, and drain-induced barrier lowering (DIBL) was calculated from the shift in threshold voltage between linear and saturation bias conditions.

$$SS = \left(\frac{d\log_{10}(I_D)}{dV_{GS}}\right)^{-1} \tag{1}$$

$$\text{DIBL} = \frac{|V_{\text{th,lin}} - V_{\text{th,sat}}|}{V_{DS,\text{sat}} - V_{DS,\text{lin}}} \tag{2}$$

The output conductance g_d was extracted from the slope of the I_D–V_{DS} curve in saturation. The on-current (I_{on}) and off-current (I_{off}) were extracted from the I_D–V_{GS} characteristics

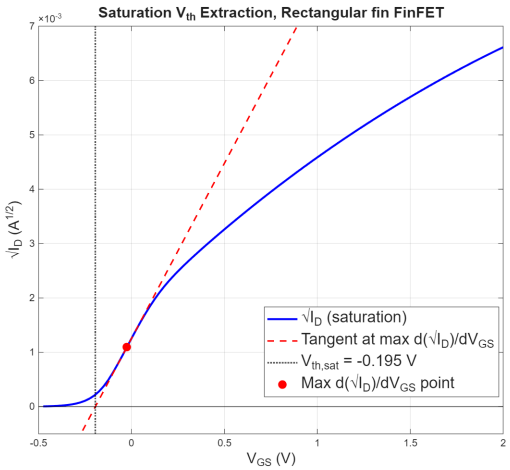


Fig. 7: Saturation V_{th} Extraction

by selecting the current values at gate voltages closest to the defined off-state ($V_{GS} \approx 0\,\mathrm{V}$) and on-state ($V_{GS} = V_{DD}$) under saturation conditions.

The on-current (I_{on}) and off-current (I_{off}) were obtained by identifying the indices that minimise $|V_{GS} - V_{\mathrm{on}}|$ and $|V_{GS} - V_{\mathrm{off}}|$, respectively, and extracting the corresponding drain current values from the I_D–V_{GS} data. The on- and off-state gate voltages were defined as $V_{\mathrm{on}} = V_{DD} + V_{\mathrm{th}} - |V_{\mathrm{th}}|$ and $V_{\mathrm{off}} = V_{\mathrm{th}} - |V_{\mathrm{th}}|$, respectively.

III. Discussion & Comparison of DC Parameters

Parameter	Rectangular Fin	Rounded Fin	NS GAAFET
$g_{m,\mathrm{lin,max}}$ (μS)	4.82	3.01	10.50
$g_{m,\mathrm{sat,max}}$ (μS)	23.08	21.18	32.51
$V_{\mathrm{th,lin}}$ (V)	-0.047	-0.089	0.500
$V_{\mathrm{th,sat}}$ (V)	-0.195	-0.350	0.461
SS (mV/dec)	83.21	83.04	60.13
DIBL (mV/V)	106.59	187.57	28.40
g_d (μS)	2.77	3.60	4.15
I_{ON} (μA)	21.04	16.83	24.24
I_{OFF} (A)	1.99×10^{-10}	1.70×10^{-9}	2.24×10^{-15}
$I_{\mathrm{ON}}/I_{\mathrm{OFF}}$	1.06×10^{5}	9.92×10^{3}	1.08×10^{10}

TABLE I: Extracted DC parameters for FinFET with different fins, and dual NS GAAFET

A. Transfer Characteristics (Id–Vg Analysis)

1) ***FinFETs–Rectangular vs Rounded Fin:*** The rectangular fin comes out ahead at 4.82 µS vs 3.01 µS (a drop of around 37%) for the rounded fin, for which the total cross-section available for current conduction shrinks. The gate has less material to modulate, so gm falls. The same trend holds in saturation, but the gap narrows considerably to about 8%. The channel is already pinched off near the drain, so the current is less sensitive to the exact cross-sectional geometry and more dominated by the charge at the source.

Both threshold voltages shift more negatively for the rounded fin. Rounding removes the sharp top corners, reducing the

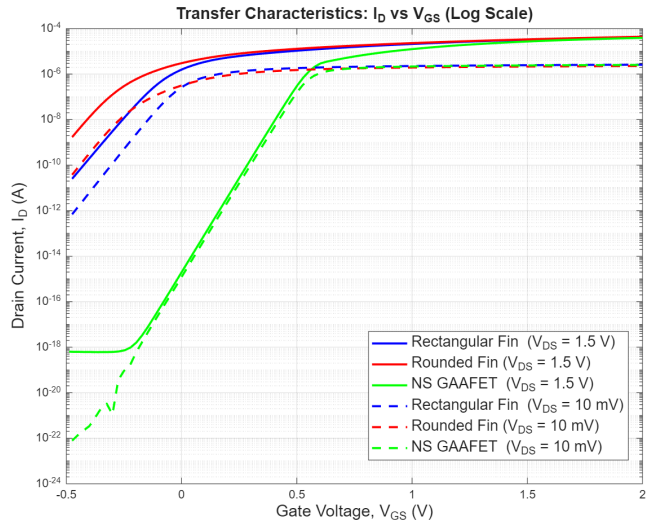


Fig. 8: Transfer Characteristics (Id–Vg) (Log Scale)

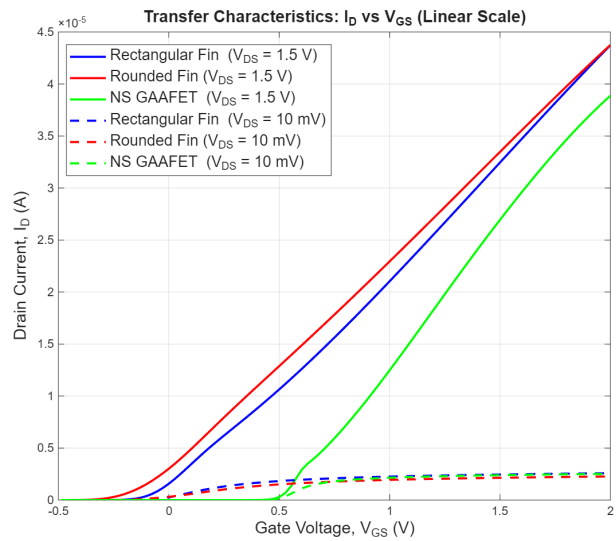


Fig. 9: Transfer Characteristics (Id–Vg) (Linear Scale)

effective flat-gate area at the top of the fin and weakening the gate's electrostatic hold over the channel. With less gate coupling opposing the drain field, the drain more effectively lowers the source-side barrier. This effect is more pronounced in saturation due to a stronger drain-induced electrostatic influence, significantly lowering the channel barrier, hence the higher DIBL is for the rounded fin. Also, rounding reduces the effective flat-gate area so with less gate area fighting the drain field. It is also noted that the DIBL values for both fin structures exceed the typical acceptable limit of $\sim 100\,\mathrm{mV/V}$, indicating pronounced short-channel effects in both cases, with the rounded fin exhibiting a more severe degradation.

Rounding reduces field non-uniformity at the fin corners, leading to a more evenly distributed potential along the channel and reducing leakage pathways. Thus, the channel turns off more efficiently, giving a steeper subthreshold slope and a slightly lower SS in the rounded fin is observed.

The rectangular fin exhibits a higher I_{on} compared to the rounded fin due to its larger conducting cross-section and greater gate-controlled channel area, which together support higher carrier density and increased drive current. In contrast, the rounded fin shows a higher I_{off}, because of more leakage current in the off-state. Consequently, the rectangular fin achieves a significantly higher $I_{\text{on}}/I_{\text{off}}$ ratio, indicating better switching performance.

*2) **FinFET vs GAAFET**:* The GAAFET demonstrates significantly improved electrostatic control compared to the rectangular FinFET, reflected in its positive shift in threshold voltage and much steeper subthreshold swing. The gate fully surrounds the channel, minimising electrostatic coupling leakages and maximising gate-to-channel capacitance to ensure uniform electrostatic control across the entire channel cross-section, resulting in stronger modulation of carrier density.

The FinFETs sit at negative threshold voltages, meaning they are depletion-mode devices that are already on at zero gate bias. The GAAFET has positive threshold voltages, making it an enhancement-mode device that is off at Vgs = 0. This difference comes entirely from the gate work function. The FinFETs use n^+ polysilicon gate whose work function depends on doping and shifts toward the silicon conduction band. The difference between the gate work function and the semiconductor work function, ϕ_{ms} is close to 0 so there is little built-in electric field at the gate, not enough to push carriers out of the channel. So the energy bands in silicon don't bend enough to form a depletion region thus the channel remains undepleted at zero bias. The GAAFET uses TiN with a tunable work function of 4.1–5.3eV. ϕ_{ms} becomes larger and creates a stronger built-in electric field which shifts the bands more strongly even at V_{gs} = 0, to push carriers away from the channel which becomes depleted.

This enhanced control also results in a substantially lower subthreshold swing, approaching the theoretical minimum thermodynamic limit of $\sim 60\,\text{mV/dec}$ at room temperature, indicating near-ideal switching behaviour. SS is set by how sharply the gate can switch the channel from off to on in weak inversion, which comes down to the ratio of gate-to-channel capacitance vs the total capacitance the channel sees. For the FinFET, the gate doesn't fully surround the channel, so there are depletion regions and fringing fields that reduce gate efficiency, pushing SS above the ideal.

The GAAFET exhibiting orders of magnitude lower I_{off} than the finFET due to all three mechanisms (higher Vth, lower SS and DIBL) suppressing off-state leakage to near-zero simultaneously. In a real fabricated device, defects and interface traps would raise this floor. Its I_{on} is slightly increased due to improved carrier confinement and more effective channel modulation. Consequently, the $I_{\text{on}}/I_{\text{off}}$ ratio is significantly enhanced, indicating superior switching performance. This enormous gap is almost entirely driven by the Ioff side of the ratio. For digital logic, this ratio determines how well a transistor can distinguish between a logic 0 and 1 state, and how much static power the circuit burns when idle. A ratio of 10^{10} means essentially zero static leakage in simulation, which is the ideal target for any low-power application.

The reduced DIBL highlights the advantage of the gate-all-around architecture in suppressing short-channel effects compared to the FinFET. DIBL happens when the drain field penetrates through the channel and lowers the source-side potential barrier so the drain control is significant to the gate control. In the GAAFET, every point of the channel cross-section is maximally close to a gate surface with no uncontrolled face, so the gate field dominates everywhere and the drain field struggles to reach the source end.

g_m is also enhanced in the GAAFET in both linear and saturation regions. A higher g_m indicates that small changes in gate voltage produce larger changes in drain current. This improvement in g_m directly translates to higher intrinsic gain and better analogue performance.

*3) **Id–Vg Curve Analysis**:* From the transfer characteristics in Figure 8 and 9, clear differences in switching behaviour are observed among the device structures.

In Figure 8, the most striking visual feature is how far right the GAAFET's curves are shifted compared to both FinFETs. The green curves don't begin rising until around $V_{gs} = 0.5\,\text{V}$, while the FinFETs are already conducting significantly at negative gate voltages. The GAAFET's subthreshold region (the rising portion of the dashed green curve) is also visibly steeper than both FinFETs, which indicates the near-ideal $60\,\text{mV/dec}$ SS. The larger vertical separation between the solid and dashed curves for each device at low V_{gs} is a visual measure of DIBL as the saturation curve sits higher above the linear curve in the off-state, indicating more drain-induced leakage. The FinFETs show a clear separation between their solid and dashed curves even at $V_{gs} = -0.5\,\text{V}$, whereas the GAAFET's solid and dashed green lines are essentially on top of each other until well below threshold, confirming its much lower DIBL of $28.4\,\text{mV/V}$.

The GAAFET also reaches a comparable on-state current to the FinFETs at high V_{gs} despite its lower overdrive, showing the GAAFET's superior gate efficiency compensating for its higher V_{th} once strongly on.

In Figure 9, the slope of the saturation curve directly represents g_m. The rounded fin has a steeper slope than the rectangular fin at high V_{gs} despite having lower $g_{m,\text{max}}$. This is because the rounded fin's lower V_{th} means it reaches a given overdrive earlier, so at the same absolute V_{gs} it has been on longer and the curve appears steeper. Thus, comparing devices at fixed V_{gs} rather than fixed overdrive can be misleading. The GAAFET starts rising much later due to its higher V_{th} but then climbs with a steeper slope than both finFETs once it turns on, visually demonstrating its higher g_m.

The linear region curves at $V_{ds} = 10\,\text{mV}$ are more flat and near zero current, regardless of V_{gs} as in the linear region, current scales directly with V_{ds}. At negative V_{gs}, the FinFETs show a visible solid curve ($V_{ds} = 1.5\,\text{V}$) sitting above the dashed curve ($V_{ds} = 10\,\text{mV}$) meaning current is still flowing in saturation indicating depletion-mode behaviour. As at negative V_{gs}, the saturation current is higher than the linear current

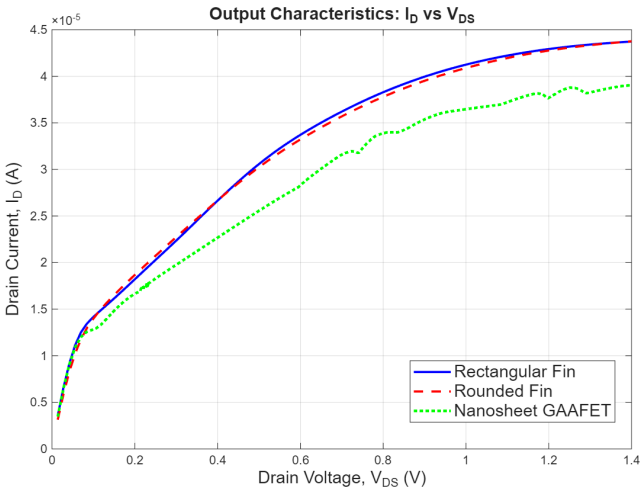


Fig. 10: Output Characteristics (Id–Vd)

shows the drain voltage is helping keep the channel on, which is attributed to DIBL occurring.

B. Output Characteristics (Id–Vd Analysis)

*1) **FinFETs–Rectangular vs Rounded Fin**:* A higher g_d in the rounded fin indicates poorer saturation behaviour, as the drain current continues to increase with V_{DS} instead of remaining constant. The reduced corner-induced field strength allows the drain potential to more strongly modulate the channel, leading to less ideal saturation behaviour.

In Figure 10, the two FinFETs reach saturation very gradually with no clear pinch-off. This is a direct consequence of their high DIBL (107–188 mV/V): the drain field continuously lowers the source barrier as V_{ds} increases, preventing the channel from properly pinching off and keeping the current rising. The rectangular and rounded fins are nearly similar on this plot, which makes sense given their similar Ion values.

Analytical doping in the fin creates a spatially varying dopant distribution, typically peaking near the sidewalls 5e19 and decaying into the centre of the channel, 1e17. This is used to model source/drain diffusing into the channel for a realistic pn junction formation between source/drain and fin as the heavily doped edges behave more like S/D extensions. In the rounded fin, a visible doping gradient is observed due to increased overlap of the source/drain Gaussian profiles within the reduced silicon volume, resulting in a smoother spatial variation in concentration. In contrast, in the rectangular fin, the sidewall regions are abruptly dominated by the high source/drain doping, leading to a sharp junction with little visible gradient as seen in Figure 1.

*2) **FinFET vs GAAFET**:* The GAAFET's g_d is higher at $4.15\,\mu$S vs $2.77\,\mu$S for the rectangular FinFET, which is unexpected given its much lower DIBL. This is because g_d is being extracted as the last point of the Id–Vds sweep, which corresponds to the maximum Vds but in saturation, the GAAFET has a V_{th} of $+0.461$ V while the FinFET sits at -0.195 V. That means at the same V_{GS}=2V, the FinFET is operating at ~ 2.195 V overdrive while the GAAFET is only at ~ 1.539 V overdrive. So the FinFET is operating at $\sim 43\,\%$ higher overdrive than the GAAFET when gd is extracted. A higher overdrive pushes the finFET deeper into saturation, flattening the Id–Vds curve and lowering gd. The GAAFET being at lower overdrive means its curve is less flat at that point, directly inflating its extracted gd. The GAAFET, despite better intrinsic electrostatics, is being evaluated at a relatively lower overdrive where saturation is less complete and the curve still has more slope.

In Figure 10, the GAAFET sits much lower in absolute current across the whole sweep. This is because of the lower overdrive of the GAAFET compared to the FinFET which is simply much more strongly turned on at that bias point, driving more current despite the GAAFET's superior gate efficiency. Despite the lower current level, its curve starts to flatten in the high-Vds region compared to the FinFETs, which is consistent with its far superior DIBL so the gate retains control against the drain field much more effectively, so the current saturates more cleanly.

There are visible step-like discontinuities in the GAAFET output curve, and this is because the two stacked nanosheets do not enter saturation simultaneously. The lower sheet sits closer to the substrate and is partially electrostatically screened from the gate by the upper sheet above it. This means the gate couples slightly differently to each sheet, so they each have a marginally different effective V_{th} and V_d, sat. As V_{ds} sweeps up, the upper sheet pinches off first then the lower sheet pinches off at a slightly higher V_{ds}. Each pinch-off event flattens the contribution from that sheet, producing the staircase-like shape rather than a smooth saturation curve. This is attributed to TCAD simulation of multi-sheet stacked GAA structures and would be smoothed out in a real device by process variation and inter-sheet coupling.

IV. Conclusion

This report demonstrates the progressive improvement in transistor electrostatic performance achievable through architectural evolution from FinFET to gate-all-around nanosheet FET. Fin corner rounding, while unavoidable in practical fabrication, consistently degrades device performance relative to the ideal rectangular fin. The reduced gate-controlled cross-section and weakened top-gate coupling lead to higher DIBL, increased leakage, and a degraded $I_{\text{on}}/I_{\text{off}}$, despite the minor improvement in SS.

The dual NS GAAFET outperforms both FinFET variants across nearly all metrics. The adoption of a TiN metal gate corrects the threshold voltage to a physically appropriate positive value, making the GAAFET a proper enhancement-mode device unlike the depletion-mode FinFETs resulting from the n^+ polysilicon gate work function. This, combined with the superior electrostatic confinement of the gate-all-around, yields an $I_{\text{on}}/I_{\text{off}}$ ratio of 1.08×10^{10}, demonstrating the compelling case for NS GAA architectures as the transistor technology of choice for low-power, high-performance applications at advanced nodes.